Air Quality Sensor Data Analysis

COMP5048 Assignment 2

line 1: Given Name Surname
line 2: *Student ID number*

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line 4: Your tutorial group number
line 5: Your tutor's name

I. Task 1 Overall Pattern Discovery

We analysed timestamp, CO(GT), C6H6(GT), NOx(GT), NO2(GT), PT08.S1(CO), PT08.S5(O3), temperature and relative humidity. The value -200 was treated as missing without imputation; 6,941 complete hourly records were retained from 9,357 records.

A. Visualisation

The CO-C6H6 scatterplot identifies concentration regions and seasonal variation. The daily heatmap shows changes across the eight attributes, while parallel coordinates use a common 0-1 scale to compare pollutants, sensor responses and environmental conditions.

B. Discovery

Two candidate profiles are visible. The sparse high-pollution region (CO >= 3 and C6H6 >= 20) has median CO 4.7, C6H6 24.2, NOx 536 and NO2 159. The dense low-pollution region (CO <= 1 and C6H6 <= 5) has medians of 0.7, 2.7, 67 and 60. Both sensor responses shift in the same direction.

The high profile contains 771 hours (11.1%), mainly in autumn and 17:00-20:00. The low profile contains 1,323 hours (19.1%), mainly from 00:00-06:00.

C. Limitation

The views show high- and low-pollution patterns, but the relationship is continuous. The boundaries are interpretive only; no clustering algorithm was used.